



IARC warning

The WHO warning of red meat and processed meat being possibly carcinogenic to human beings. <https://digit.in/jan20-33>



Red meat recommendations

The paper based on a survey of studies, finding no link between meat consumption and cancer. <https://digit.in/jan20-34>

cereals and millets can provide most of the required amino acids. It does note that an excess of red meat, along with organ meat and eggs can cause obesity, diabetes, cardiovascular disease and cancer because of its high cholesterol content. While noting that in India, it is important to increase consumption of animal based foods such as eggs, milk and meat. The only kind of meat that the guideline says to outright avoid is organ meats such as liver, brain and kidney. The guide positively recommends consumption of eggs, even though egg yolk has cholesterol. In all, considering the extent to which plant based foods are consumed in India, there is no focus on the need to reduce the consumption of meat in diets.

Perhaps the most alarming call for the avoidance of red meat was from the International Agency for Research on Cancer, which is an intergovernmental body forming part of the World Health Organisation of the United Nations. In 2015, the IARC announced that red meat was probably carcinogenic to humans, noting that the associations were observed most strongly for colorectal cancer, but also seen for pancreatic cancer and prostate cancer. Additionally, the announcement also declared that processed meat was carcinogenic to humans. A panel of experts concluded that the chances of getting colorectal cancer increases by as much as 18 percent, when only 50 grams of processed meat is consumed every day. At that time, the consensus seemed to be clear, red meat and processed meat were linked to both heart disease and cancer. It was better to reduce the consumption of red meat.

In November 2019, a paper was published in the Annals of Internal Medicine, that seemed to go against popular wisdom. The report caused a stir, and was reported around the world. The paper conducted a systematic survey of five studies, and concluded that there was no link between consumption of red meat or processed meat, and increased mortality. Basically, the supposed negative health effects of eating red meat and processed

meat was negligible. The paper was more of a meta commentary on the previous nutrition studies that linked the consumption of meat with negative health effects. It said that the data was not very well collected, and a clear correlation was not established. According to the researchers, those who want to can continue to consume red meat and processed meat, without facing any negative health consequences.

The researchers themselves note some problems with their findings, such as the fact that the difference in meat consumption among the people included in the study was not much. Academics have identified some of the problems with the new study as well. This is because the study does not distinguish between the different ways in which meat is prepared for consumption. While cooking and boiling are considered safe, grilling or broiling are known to cause heterocyclic amines or HCAs, which are believed to be carcinogenic. The paper never defines what a portion of meat is. The paper also excludes studies conducted on fewer than 1,000 people, and for a period shorter than six months. This actually excludes a large number of studies, and the researchers do not adequately explain this. While the new paper applied the results to the entire population, dieticians base their recommendations on a number of factors such as medication, age, caloric needs, as well as personal and cultural preferences. The paper also contradicts the results of the studies that it is based on. Researchers from the University of Newcastle, Pennsylvania State University and Harvard have all publicly noted that the recommendation of the new paper should be taken with a dollops of salt.

NUTRITIONAL SCIENCE

While the findings of the new paper may be contentious, it does highlight some of the problems in nutritional science. The problem here may well be in how these studies are conducted, which mean that it might never be possible to get a definitive answer. The

bodies of those who follow vegetarian diets and non vegetarian diets are very different. It is very difficult to study for long durations of time, the effect of consuming and not consuming meat for the same person. You cannot just pick random groups of people, and give them certain diets to stick to, and observe what conditions they are affected by. You also cannot give a meat placebo to a control group, to compare between groups who thought they have eaten meat, and groups that have actually eaten meat. To make sure that say a control group eats only plant based food is unethical in itself. Most of nutritional science is based on observational studies instead of random testing. What this means is that people with a particular diet or lifestyle are tracked over long periods of time, to see what kind of symptoms or conditions they develop. However, there might be other factors affecting the underlying diet, that contribute to these conditions - say diabetes or cancer, which have nothing to do with the food. Just eating meat and not eating meat is rarely the only difference between the two groups. For example, vegans, who take a lot of trouble deciding on their diet, might also be more health conscious than the average meat eater. Even within meat eaters, you might have groups that eat healthy, and others who eat a lot of junk food, and both are just classified as meat eaters. Thus, such studies are not without their flaws, and the researchers themselves are most likely to tell you the limitations of their results. Additionally, the thing that makes nutritional findings really dubious is that the labs that conduct these research are funded by collaborations of multinational companies that have an interest in selling people what they eat. These can range from sugar water to exported waste meat products. Because of where the money is coming from, the science can be coloured or selectively interpreted to favour these companies that have an interest in selling food. This goes to show why the question is incredibly difficult, if not impossible to answer. **4**